Kalp Shah

Roll No.:24110152
B.Tech - Electrical Engineering
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Education

Degree	Institute	CGPA/Percentage	Year
B.Tech.	Indian Institute of Technology, Gandhinagar	8.67/10 (Current)	2024-2028

Experience

- **AIResQ ClimSols** — *May 2026 - Jul. 2026*
  Machine Learning and Computer Vision Research Intern — Gandhinagar, India
  - Engineered a **custom Camera and Radar Device** for collection and validation of flood depth data
  - Implemented **Single View Metrology** CV algorithms to measure flood depth from CCTV cameras
  - Trained **YOLOv8** Models to detect flood level markers painted on light poles with **mAP@50 of 0.78**
  - Extracted features like flood waterline using **semantic segmentation** with DeepLabV3+
- **ABSix Robotics** — *May. 2025 - Jul. 2025*
  SWE Intern — Gandhinagar, India
  - Engineered a **cross-platform desktop application (Python, Qt)** with real-time robot-student communication via the **Dynamixel SDK** deployed across 15+ challenges and actively used by **100+ students**
  - Integrated a **RAG pipeline (LangChain + ChromaDB)** into the platform for students to query robotics concepts

Projects

- **Limit Orderbook and Trade Matching Engine**
  Personal Project — Github
  - Built a high-performance Limit Order Book and trade matching engine in **C++** with support for **multiple order types** including Market, Limit, Fill-or-Kill (FOK), and Immediate-or-Cancel (IOC) orders
  - Implemented low-latency price-time priority matching to support **4000 Transactions per second**
  - Optimized using **cache-friendly** data layouts, **lock-free design**, and **deterministic event ordering**
- **Concurrent Asynchronous Logging System**
  Personal Project — Github
  - Implemented an asynchronous logging library using **C++** enabling multiple threads to log concurrently
  - Developed a dedicated background writer thread with a **thread-safe queue** using std::mutex, std::condition_variable, and batched log flushing to reduce synchronization overhead
  - Benchmarked with **128 concurrent threads** at **1.7 million messages/sec** with ordered, thread-safe output
- **Realtime Carbon Credits Market Analysis Platform** — *Nov. 2025 - Dec. 2025*
  Inter-IIT Tech Meet 14.0 - Pathway, High Preparation Problem Statement — Github
  - Architected a full-stack real-time analytics platform (**Next.js, WebSocket, PostgreSQL**) serving live carbon credit pricing, trend signals, and risk analysis to end users as part of an **8-person team**
  - Built a data streaming pipeline using **Kafka** and **Debezium** (CDC) to capture live **PostgreSQL** changes, cutting data freshness latency from **60 minutes to under 3 seconds**
  - Developed a multi-agent **RAG system (LangChain, DeepSeek)** with role-separated agents for retrieval, reasoning, and synthesis generating context-aware investment recommendations
- **Landing Location Detection Pipeline for Mars Rover using Classical Computer Vision** — *Jan. 2026 - Apr. 2026*
  Prof. Shanmuganathan Raman, IITGn — Github | Research Poster
  - Developed a Computer Vision pipeline to identify **safe landing zones on Mars** with **ISRO** under compute constraints
  - Designed a modular system utilizing Sobel derivatives for **slope detection**, Fast Fourier Transforms for **texture analysis**, and the Maximal Rectangle Algorithm for **zone identification**
  - Achieved a low computational power requirement solution with **1.5 seconds runtime** on consumer-level hardware

Achievements

- **Selected to Barclays Mentorship Programme India**, Learned capital markets; built an FX hedging backtester *2026*
- **Expert on Codeforces**, Handle: *kalpshah1808*, **India Rank ∼450** out of 98000 *2026*
- **Integral Cup Round 1 Rank 47**, among 2500+ participants, by **Optiver** and **Da Vinci Trading** *2026*
- **AMS Derive Round 1 Rank 150 and Round 2 Rank 17**, among 2500+ participants, by **Jane Street** and **QRT** *2026*
- **JEE Advanced 2024 AIR 6637 JEE Mains 2024 AIR 3917**, among 1.4 million candidates *2024*

Skills

- **Languages**: Python, C/C++, SQL, HTML/CSS, Javascript, Assembly, Verilog
- **ML/CV**: PyTorch, OpenCV, scikit-learn, Langchain, Pandas, Numpy, Matplotlib
- **Backend/Tools**: Node.js, WebSocket, PostgreSQL, Kafka, Debezium, Docker, REST APIs, Git, CMake
- **Frontend**: React.js, Next.js, Tailwind CSS